Capstone Deliverable 4: Analysis

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Project Overview

Canada has committed to hit zero-emission vehicle (ZEV) adoption targets of new vehicle sales being 20% by 2026, 60% by 2030, and 100% by 2035. Although recent declines in battery electric vehicle (BEV) adoption indicates some barriers on the consumer side surrounding charging infrastructure availability. This project will address the gaps by providing insights to support decision making around EV charging infrastructure.

The client for this project is Natural Resources Canada (NRCan) with their Zero Emission Vehicle Infrastructure Program (ZEVIP). This analysis with provide NRCan with the evidence to determine where infrastructure gaps exist and how many chargers will be needed under adoption scenarios. The objective of this analysis is to identify infrastructure gaps and forecast future demand.

Data Overview

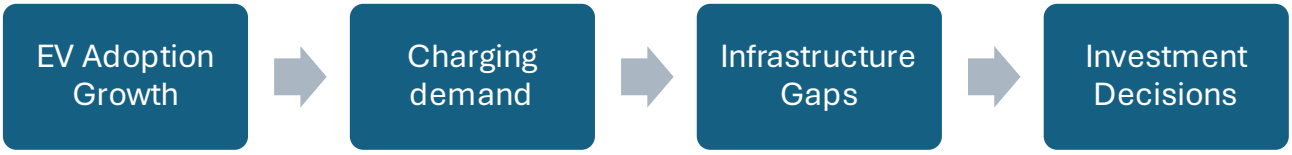
The datasets used in this analysis are all publicly available with some being from official sources like the government of Canada. The main datasets used is the census performed in 2024, alternative fuel stations, ZEV registrations, and highway systems. The census and alternative fuel stations datasets provide information to support overlaying population with charging stations to find infrastructure gaps. The highway system dataset is a GDB file type. This type is a collection of point and lines to visualize where major highway systems are.

Data Dictionary

Field	Type	Description
Latitude	Float	Provides geographical coordinate of where the charger is located
Longitude	Float	Provides geographical coordinate of where the charger is located
PowerKW	Numeric	Information of the charger's maximum power output
ChargerType	Categorical	The type of charger type either level 2 or DC fast charger
EV Network	String	Information if the charging station is networked or not. Either not networked or networked through Chargepoint or Tesla etc.
ZEV_Sales	Integer	The number of ZEV registration and sales recorded quarterly.

Open Date	Date	Present in the charge station dataset to show when stations were opened. Can be used to show growth of charging stations.
Population	Integer	Provided though the census in 2024 to show the population in certain areas around Canada.

Conceptual Model Diagram



To meet the government of Canada's EV sales, target the overall adoption of EV's will also grow to meet the target. As adoption grows so does the demand for more charging stations to support the drivers. The more demand there is for charging stations the need for better infrastructure needs to be addressed. The infrastructure gaps in this project are the number of public charging stations available as well as the power needed to maintain a stable power grid. These end up leading to the decisions needed to be made on where the investments should be focused. Overall, this process will help address the need to meet the EV adoption targets set by the government.

Solution Build

The solution build for this project resulted in a decision-support analytics product that helps stakeholders identify EV charging infrastructure gaps and plan cost-effective expansions across Canada. It combines geospatial visualization, demand forecasting, and scenario-based cost/utility evaluation into one coherent planning workflow intended to support funding and siting decisions for NRCan's infrastructure allocation.

The product consists of (1) a Canada-wide infrastructure-gap map and (2) an analytical model suite that estimates how many charging ports are required under different EV adoption and fleet-mix assumptions, then evaluates investment value across charger types. The geospatial component overlays three layers—National Highway System routes, population density constrained to inhabited ecumene areas, and charging station point locations—so infrastructure shortfalls can be interpreted in terms of both where Canadians live and where they drive. The forecasting component uses recent registration trends and scenario planning (different BEV/PHEV splits) to translate projected ZEV growth into required public charging ports, highlighting how charger needs vary substantially depending on the fleet composition. The value assessment component uses a Monte Carlo simulation approach (1,000 runs with a fixed budget allocated between Level 2 and DCFC) to compare the utility trade-offs of deploying many lower-cost Level 2 chargers versus fewer, higher-

impact DC fast chargers needed for longer-distance travel. Outputs from these components are designed to support recommendations such as prioritizing underserved regions and incomplete corridors, planning infrastructure by fleet composition, and balancing near-term cost-effectiveness with long-term ZEV objectives.

Analysis & Key Findings

The infrastructure gap analysis was built as a multi-layer national map that overlays (1) the National Highway System, (2) population density filtered to inhabited ecumene areas, and (3) charging station locations to identify where chargers are missing relative to both resident demand and travel corridors. This design choice is important because it avoids placing chargers only where population is highest, and instead highlights gaps that affect long-distance travel and inter-city accessibility along highways.

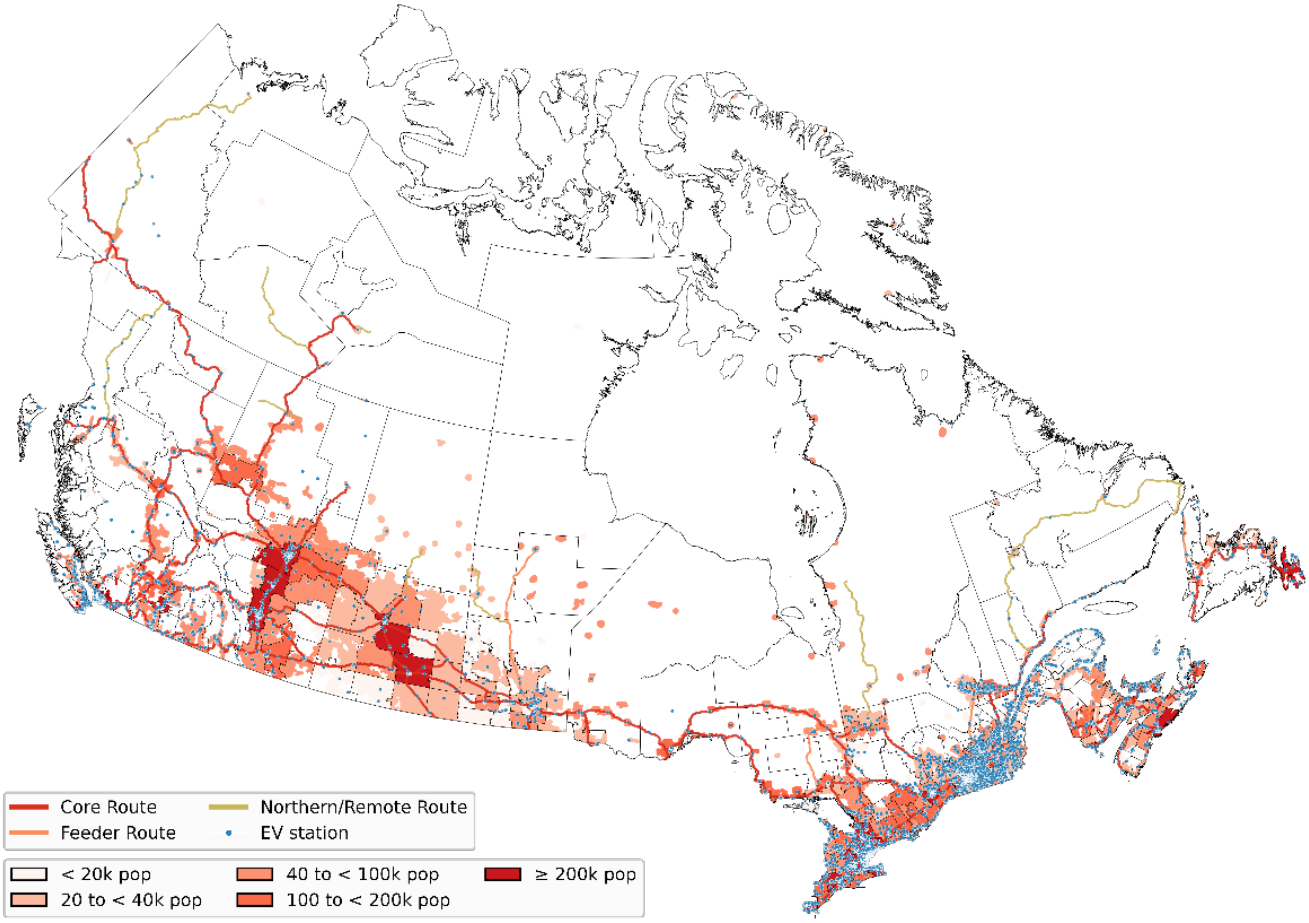


Figure 1: Federal infrastructure gaps map

The spatial results show charging infrastructure is heavily concentrated in Québec and Ontario (notably along the Québec–Windsor corridor) and in B.C., while much of the rest of Canada shows limited coverage outside major urban "hotspots" (Vancouver, Calgary, Edmonton, Winnipeg). Québec appears as a clear leader in readiness: the project notes that Québec accounts for over 40% of Canada's new ZEV registrations and has about 35 EVs per public charger, reinforcing the interpretation that early infrastructure build-out supports adoption momentum. Perceived availability also aligns with this pattern, with reports that 46% of EV owners in Québec find station availability adequate versus 28% in Ontario and 21% in B.C., suggesting that even provinces with relatively better coverage can still experience adequacy gaps depending on demand intensity and user expectations.

Canadian EV owners "overwhelmingly satisfied" with EVs, access to charging a key purchase factor

Figure 2: Electric Autonomy Canada Headline regarding charging infrastructure satisfaction

To translate adoption into infrastructure requirements, the demand model used differentiated public-charger intensity assumptions by powertrain type (with BEVs requiring more public chargers

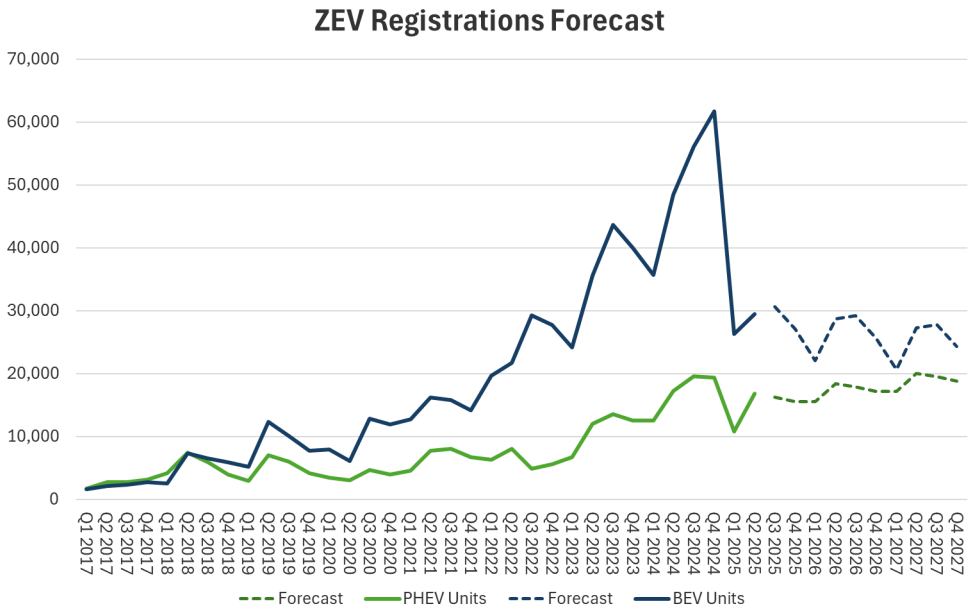


Figure 3: ZEV registrations forecast chart

per 1,000 vehicles than PHEVs) and then applied scenario planning to test how different BEV/PHEV mixes change the number of ports needed. Using the chosen planning assumptions (including a BEV-to-PHEV trend scenario and alternative 80/20 splits), the result was an estimate of 15,218 required new charging ports over the next two years under the stated conditions, described as roughly a 40% increase over existing infrastructure.

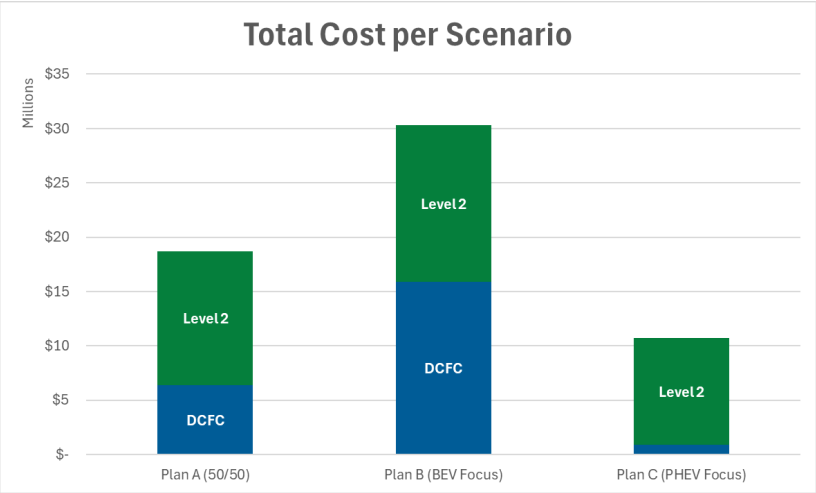


Figure 4: Cost Difference per Scenario

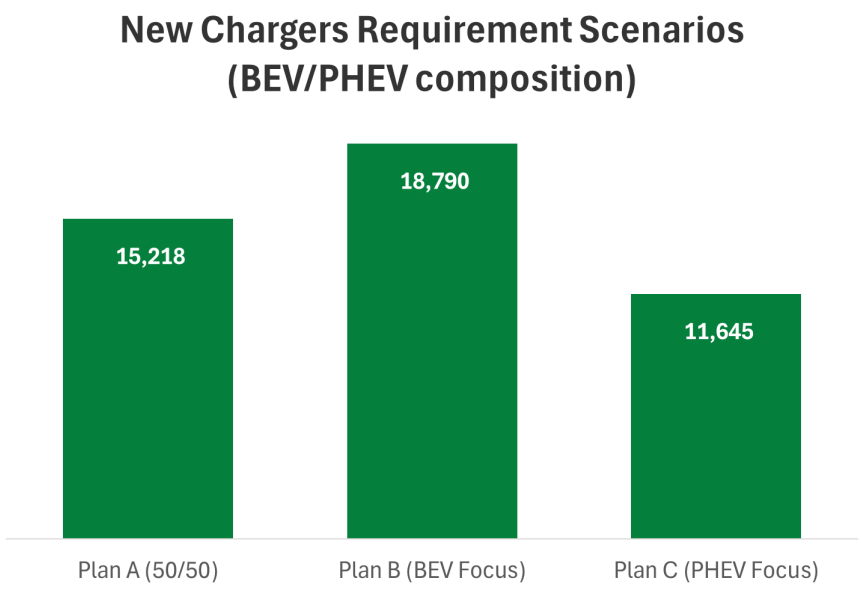


Figure 5: New charger requirement scenarios

For investment value, the project evaluated the trade-off between installing more Level 2 ports (lower unit cost, higher quantity) and fewer DC fast chargers (higher unit cost, higher per-unit utility for long-distance travel) using both scenario cost tables and a Monte Carlo simulation. The Monte

Carlo model ran 1,000 simulations with a $100M budget, sampling Level 2 costs between $5k–$12k and DCFC costs between $80k–$150k, and applied utility scoring (1 point for Level 2 and 8 points for DCFC in the parameter sheet, with additional comparisons explored). The scenario summary emphasizes that a balanced approach can preserve most of the utility benefits of a DCFC-heavy strategy at a much lower share of total spending, and it also flags that a PHEV-focused strategy may look attractive short-term but risks misalignment with long-term full zero-emissions goals if DCFC buildout is deferred.

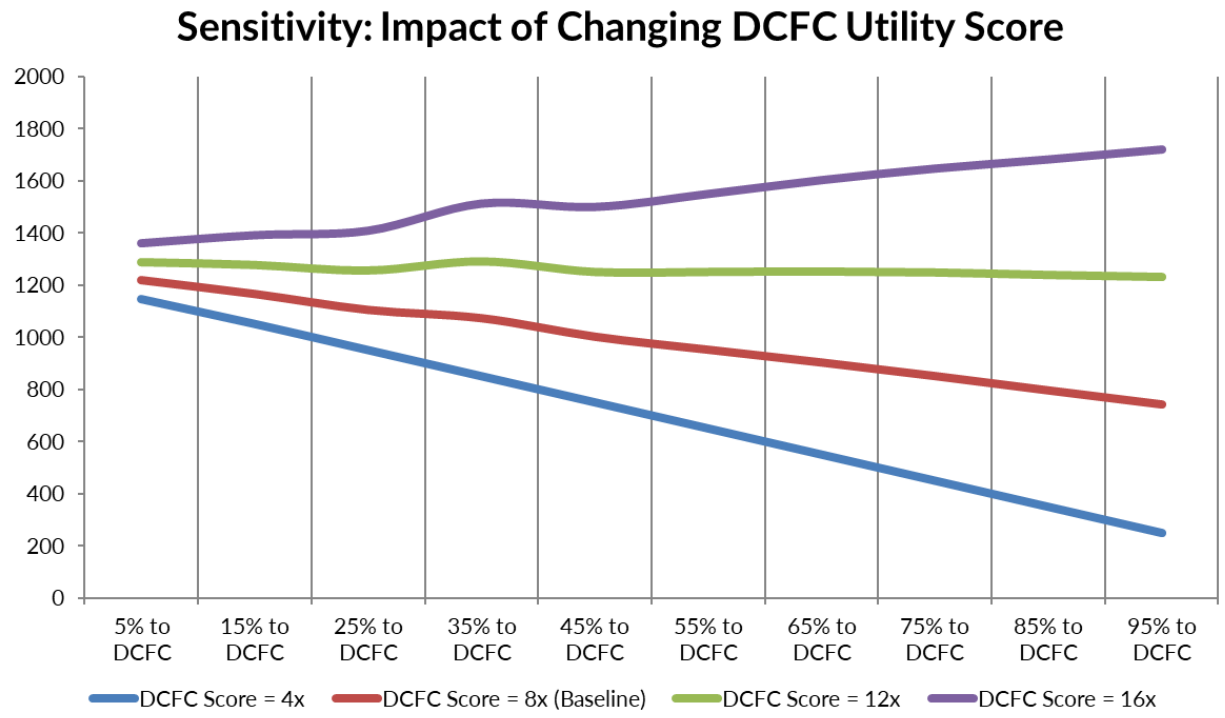


Figure 6: Monte Carlo simulation plot

Changes from Proposal

The main change from the original proposal was the scope of the analysis. Originally the scope was to analyze ZEV's but later decided to focus only on EV's as there was very limited amount of data present on other ZEV's other than EV's. Hydrogen cell vehicles were the other present type of ZEV however this would be significantly different from the EV analysis as hydrogen cells use a different type of refueling station. Another change from the original proposal was instead of generating a single forecast for EV adoption and demand, we decided to consider different scenarios to run forecasting. This change was due to the uncertainty of the EV market as demand and incentives can change very drastically.

Conclusion & Recommendations

The infrastructure is important to reach their target to make charging stations more accessible and stable. Currently, Canada has 35,000 charging stations across the country but will require around 100,000 chargers. There are various regions around Canada that are lacking in public charging stations and will need to allocate investments to these areas. Overall, when considering which chargers are worth more per station, the DC fast chargers provide more value over the long term despite the higher initial costs. Even though the cost for the DC fast charger is around 10 times the cost of a level 2 charger, over time these will provide more value over time.

Recommendations

Prioritizing underserved regions where there is high density of population with little coverage of charging stations. Another area that should be prioritized are the major highway systems where people are taking longer road trips and might need to recharge before leaving. Considering this, to fit the governments budget they would need a combination of DC fast chargers as well as level 2 chargers. Placing the fast chargers around major highway systems is a high priority with the level 2 chargers being placed in areas with lower priority.

References

EV sales data:

https://www.iea.org/data-and-statistics/data-tools/global-ev-data-explorer

Charging stations:

https://www.kaggle.com/datasets/tarekmasryo/global-ev-charging-stations?resource=download

Census 2024:

https://www150.statcan.gc.ca/n1/daily-quotidien/250116/dq250116b-cansim-eng.htm

Alternative fuel stations:

https://natural-resources.canada.ca/energy-efficiency/transportation-energy-efficiency/electric-charging-alternative-fuelling-stationslocator-map#/analyze?tab=location&fuel=ELEC&ev_levels=all&status=E&status=T

Highway systems:

https://open.canada.ca/data/en/dataset/c5c249c4-dea6-40a6-8fae-188a42030908